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14 November 1968
SUPERSEDING
FA-PD-ME-2492, Rev.0
24 March 1961

MILITARY SPECIFICATION
INITIATORS, CARTRIDGE ACTUATED: DELAY, M15, XM17, XM19, XM21, XM23, XM25 and XM110 METAL PARTS ASSEMBLY

1. SCOPE

1.1 This specification covers Initiators, Cartridge Actuated: Delay, M15, XM17, XM19, XM21, XM23, XM25 and XM110 Metal Parts Assembly. The Assembly will ultimately be loaded and used in conjunction with other Cartridge Actuated devices in aircrew emergency escape systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Specifications
Military
MIL-A-2550 - Ammunition and Special Weapons, General Specifications For

Standards
Military
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109 - Inspection Terms and Definitions
MIL-STD-413 - Visual Inspection Guide for Rubber O-Rings
MIL-STD-1167- Ammunition Data Card

Drawings
Munitions Command
D10520858 - Initiators, Cartridge Actuated, Delay, Metal Parts Assembly
C8846787 - Container, Ammunition, Fiber M458 for Initiators, Cartridge Actuated, Delay
C8846788 - Box, Packing, Ammunition for Initiators, Cartridge Actuated, Delay in Fiber Container M458
IEL-10520858 - Inspection Equipment List for Initiators, Cartridge Actuated, Metal Parts Assembly

FSC 1377

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(Copies of specifications, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. - The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bid or request for proposal shall apply.

ASA B46.1 - Surface Texture

(Application for copies should be addressed to the American Standards Association, 10 East 40th Street, New York, N.Y. 10016.)

3. REQUIREMENTS

3.1 General.- Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings and referenced specifications.

3.2 Physical and Mechanical.- Properties required by the drawing shall be uniform throughout the part within specified tolerance. Requirements for all alloy steel parts shall be achieved by quenching and tempering.

3.3 Hydrostatic pressure.- The initiator chamber and cap shall withstand a hydrostatic pressure of 14,000 to 15,000 pounds per square inch (PSI) for 15 seconds, minimum, without deformation of mechanical failure (see 6.2)

3.4 Shear Pin Strength.- The shear pin, after application of protective finish, shall shear when a load of 41 pounds plus or minus 5 pounds is applied at the steady rate of 0.125 inch per minute.

3.5 Protective finish.- Protective finish shall be in accordance with the applicable drawings and specifications, and shall apply to all surfaces of the component regardless of contour.

3.6 Residual Magnetism.- The metal parts assembly shall not deflect the indicator of a compass more than 3 degrees in either direction when passed in front of the compass at a distance of 6 inches.

3.7 Workmanship.- The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.7.1 Metal Defects.- The part shall be free from cracks, splits, cold shuts, inclusions, porosity or other metal defects.

3.7.2 Burr.- The part shall not have a burr which might interfere with assembly or function of the item or which might be injurious to personnel handling the item.

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3.7.3 Foreign matter.- The part or assembly shall not contain chips, dirt, grease, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.- Reference shall be made to MIL-STD-109 to define the quality assurance terms used.

4.1.2 Inspection.- Inspection shall be in accordance with MIL-A-2550.

4.2 First Article Sample

4.2.1 Initial production sample.- At the beginning of regular production a sample shall be submitted in accordance with contract requirements and shall consist of 50 unplated units of each heat treated component for hardness tests, 50 shear pins for shear pin strength, to be tested in accordance with Table I and 5 assemblies and 10 sets of component for visual and dimensional examination.

TABLE I

Paragraph Requirements	Paragraph Tests	Type of Test	Sample Size	Type of Test Dest.	Type of Test Non-Dest.
3.2	4.4.1	Hardness	50 each	X	-
3.3	4.4.2	Hydrostatic	each	-	X
3.4	4.4.3	Shear Pin Strength	50	X	-
3.5	4.4.4	Protective Finish	as specified in applicable specification	X	-
3.6	4.4.5	Residual Magnetism	each	-	X

The sample shall be manufactured using the same materials, equipment, processes and procedures as used in regular production. All parts and materials, including packaging and packing shall be the same as used for regular production and shall be obtained from the same source of supply.

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4.2.1.1 Examination and tests.- After inspection and provisional acceptance at source, the sample shall be inspected for all requirements of the drawings and specifications at a Government Laboratory or such other facility as specified in the contract.

4.2.1.2 Initial production sample failure.- Failure of the sample to comply with the requirements of the drawings and specifications shall result in sample disapproval.

4.3 Inspection provisions

4.3.1 Lot

4.3.1.1 Submission of product.- The product shall be submitted in accordance with MIL-STD-105.

4.3.1.2 Size of lot.- A lot shall consist of an identifiable group of metal part assemblies, produced by a homogeneous process.

4.3.2 Examination.- One hundred percent examination shall be performed for all critical defects. Examination for Major and Minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plan and associated criteria in table II and III. For an individual defect (single characteristics) the criteria in table II shall be used. For total defects within a major or minor group the criteria in table III shall be used. All non-conforming material shall be rejected. Inspection for rubber O-rings shall be in accordance with MIL-STD-413. The roughness comparison specimens prescribed in ASA-B46.1 shall be used as a basis of comparison for surface roughness determination. Four power (4x) magnification may be used in performing visual examination for such defect as burr and foreign matter.

TABLE II

Lot Size	Sample Size	Major		Minor	
		Accept	Reject	Accept	Reject
1-300	100% Exam.	-	-	-	-
301-1300	110	1	2	2	3
1301-3200	150	2	3	3	4
3201-8000	225	3	4	4	5
3001-22000	300	4	5	5	6

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TABLE I. Test classification

TABLE III

	Total Allowable Defects Per Sample Size			
Sample Size for Major Defects	110	150	225	300
Number of defects in class				
2 - 3	1	2	3	4
4 - 7	2	3	4	5
8 - 16	3	4	5	7
17 - 48	4	5	8	10
over 48	6	8	11	14
Sample Size for Minor Defects	110	150	225	300
Number of defects in class				
2 - 3	2	3	4	5
4 - 7	3	4	5	7
8 - 16	4	5	6	10
17 - 45	6	8	11	14
over 48	8	11	17	20

4.3.2.1 Chamber, Initiator, Drawing C8594928, covering a detail of Drawing D10520858.

Defect		Method of Inspection
CRITICAL		
None Defined.		
MAJOR		
101.	Pitch diameter of large external thread, min.	Gage
102.	Major diameter of large external thread, min.	Gage
103.	Pitch diameter of small external thread, min.	Gage
104.	Major diameter of small external thread, min.	Gage
105.	Pitch diameter of internal thread, max.	Gage
106.	Minor diameter of internal thread, max.	Gage
107.	Concentricity of datum location with pitch diameter of small external thread	Gage
108.	Concentricity of pitch diameter of large external thread with cavity	Gage
109.	Concentricity of large external thread pilot with large external thread	Gage
110.	Concentricity of counterbore with cavity including perpendicularity with base of counterbore	Gage
111.	Diameter of counterbore	Gage
112.	Diameter of large thread pilot	Gage
113.	Depth of cavity	Gage

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4.3.2.1 (Continued)

	Defect	Method of Inspection
MAJOR		
114.	Length to datum location	Gage
115.	Thickness of wall, min.	Gage
116.	Metal defective	Visual
117.	Protective finish inadequate or defective	Visual
118.	Surface finish of cavity and angle of gas port improper	Visual
119.	Burr or sharp corner	Visual
120.	Foreign matter	Visual
MINOR		
201.	Pitch diameter of large external thread, max.	Gage
202.	Pitch diameter of small external thread, max.	Gage
203.	Pitch diameter of internal thread, min.	Gage
204.	Diameter of small external thread relief	Gage
205.	Diameter of small external thread pilot	Gage
206.	Diameter of hole in gas port	Gage
207.	Diameter of cavity	Gage
208.	Diameter of chamfer and internal thread	Gage
209.	Depth of counterbore	Gage
210.	Depth of threaded hole	Gage
211.	Length of gas port	Gage
212.	Length of small thread relief	Gage
213.	Length from face of gas port to small thread	Gage
214.	Length of large external thread	Gage
215.	Length of large external thread pilot	Gage
216.	Total length	Gage
217.	Thickness of wall, max.	Gage
218.	Marking missing, incorrect or illegible	Visual
219.	Radius or chamfer missing or incorrect	Visual
220.	Surface finish improper, except as otherwise classified	Visual

4.3.2.2 Cap, Initiator, Drawing D10520557, covering a detail of Drawing D10520858.

	Defect	Method of Inspection
CRITICAL		
	None defined.	
MAJOR		
101.	Pitch diameter of large external thread, max.	Gage
102.	Minor diameter of large internal thread, max.	Gage
103.	Pitch diameter of shear pin set screw thread, max.	Gage
104.	Minor diameter of shear pin set screw thread, max.	Gage

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4.3.2.2 (Continued)

	Defect	Method of Inspection
MAJOR		
105.	Pitch diameter of small internal thread, max.	Gage
106.	Minor diameter small internal thread, max.	Gage
107.	Concentricity of sidewalls of boss cavity with pitch diameter of large internal thread	Gage
108.	Concentricity of large internal thread with midsection cavity	Gage
109.	Concentricity of firing pin hole with midsection cavity	Gage
110.	Concentricity of body, large end, with pitch diameter of large internal thread	Gage
111.	Concentricity of undercut behind large internal thread with pitch diameter of large internal thread	Gage
112.	Concentricity of body with small cavity	Gage
113.	Concentricity of pitch diameter small internal thread with small cavity	Gage
114.	Concentricity of chamfer at firing pin hole with body, large end	Gage
115.	Perpendicularity of internal boss face with pitch diameter of large internal thread	Gage
116.	Symmetry of flats with body	Gage
117.	Symmetry of thickness of mounting lugs with body, large ends	Gage
118.	Symmetry of length of mounting lugs with body, large ends	Gage
119.	Symmetry of shear pin hole with small cavity	Gage
120.	Variation of center line of mounting holes with center line of small cavity	Gage
121.	Diameter of boss cavity, max.	Gage
122.	Diameter of large thread undercut	Gage
123.	Diameter of firing pin hole	Gage
124.	Diameter of boss cavity, max.	Gage
125.	Diameter of shear pin hole, max.	Gage
126.	Diameter of small thread counterbore	Gage
127.	Diameter of chamfer, small thread	Gage
128.	Diameter of chamfer, small thread cavity	Gage
129.	Diameter of chamfer, large thread cavity	Gage
130.	Diameter of small cavity	Gage
131.	Diameter of chamfer and angle of chamfer at firing pin hole.	Gage
132.	Depth of small thread cavity counterbore, max.	Gage
133.	Depth of shear pin set screw counterbore, max.	Gage
134.	Depth of large thread cavity to face of internal boss	Gage
135.	Depth from large thread undercut to base of cartridge head recess	Gage

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4.3.2.2 (Continued)

Defect	Method of Inspection
MAJOR	
136. Length to end of effective thread, min.	Gage
137. Length to end of effective thread, max.	Gage
138. Thickness of wall, min.	Gage
139. Height of boss	Gage
140. Marking missing, incorrect or illegible	Visual
141. Metal defective	Visual
142. Protective finish inadequate or defective	Visual
143. Surface finish of small cavity undercut and mouth of large cavity and small cavity improper	Visual
144. Burr.	Visual
145. Foreign matter	Visual

4.3.2.3 Pin, Shear, Drawing B10520475, covering a detail of Drawing D10520858.

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
101. Diameter, max.	Gage
102. Total length, min.	Gage
103. Metal defective	Visual
104. Protective finish inadequate or defective	Visual
105. Surface finish improper	Visual
106. Burr	Visual
107. Foreign matter	Visual
MINOR	
201. Total length, max.	Gage
202. Profile of end	Gage

4.3.2.4 Filter, Drawing C10520414, covering a detail of Drawing D10520858.

Defect	Method of Inspection
CRITICAL	
None defined.	

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4.3.2.4 (Continued)

Defect	Method of Inspection
MAJOR	
101. Pitch diameter of thread, min.	Gage
102. Major diameter of thread, min.	Gage
103. Concentricity of thread with center hole and outside diameter	Gage
104. Symmetry of holes with outside diameter	Gage
105. Diameter of center hole	Gage
106. Diameter of holes	Gage
107. Depth of center hole	Gage
108. Depth of slot, max.	Gage
109. Location of holes	Gage
110. Angular location of holes	Gage
111. Nylon patch not properly located or fused	Manual/Visual
112. Number of holes incorrect	Visual
113. Metal defective	Visual
114. Protective finish inadequate or defective	Visual
115. Burr	Visual
116. Foreign matter	Visual
MINOR	
201. Pitch diameter of thread, max.	Gage
202. Diameter of undercut	Gage
203. Diameter of body	Gage
204. Depth of slot, min.	Gage
205. Total length	Gage
206. Length of Body	Gage
207. Width of undercut, max.	Gage
208. Width of slot	Gage
209. Location of slot	Gage
210. Marking missing, incorrect or illegible	Visual
211. Chamfer missing or incorrect	Visual
212. Surface finish improper	Visual

4.3.2.5 Pin, Firing, Drawing C10520474, covering a detail of Drawing D10520858.

Defect	Method of Inspection
CRITICAL	
None Defined.	

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4.3.2.5 (Continued)

	Defect	Method of Inspection
MAJOR		
101.	Concentricity of striker with body	Gage
102.	Concentricity of groove with body	Gage
103.	Symmetry of pin hole with body	Gage
104.	Diameter of striker	Gage
105.	Diameter of body	Gage
106.	Diameter of groove	Gage
107.	Diameter of Pin hole	Gage
108.	Depth of pin hole	Gage
109.	Length of striker, min.	Gage
110.	Location of pin hole	Gage
111.	Metal defective	Visual
112.	Protective finish inadequate or defective	Visual
113.	Surface finish of body improper	Visual
114.	Burr or sharp corner	Visual
115.	Foreign matter	Visual
MINOR		
201.	Depth of slot	Gage
202.	Width of slot	Gage
203.	Width of groove	Gage
204.	Total length	Gage
205.	Length of striker, max.	Gage
206.	Location of groove	Gage
207.	Location of slot	Gage
208.	Angle on striker end of body	Visual
209.	Radius or chamfer missing or incorrect	Visual
210.	Surface finish improper, except as otherwise classified	Visual

4.3.2.6 Assembly, Drawing D10520858, less all drawings except D10520557, C10520474 and MS29513-9

	Defect	Method of Inspection
CRITICAL		
1.	Firing pin protrusion	Gage
2.	O-ring missing	Visual
MAJOR		
101.	Burr	Visual
102.	Foreign matter	Visual
MINOR		
	None defined.	

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4.3.2.7 Assembly Drawing D10530858.

Defect	Method of Inspection
CRITICAL	
None defined.	
MAJOR	
101. Any part missing, except as otherwise classified	Visual
102. Improper assembly	Visual
103. O-ring not lubricated	Visual
104. Marking missing, incorrect or illegible	Visual
MINOR	
201. Shipping cap or plug missing	Visual
202. Protective finish inadequate or defective	Visual

4.3.2.8 Fiber Container Drawing C8846787

Defect	Method of Inspection
CRITICAL	
None defined.	
MAJOR	
None defined.	
MINOR	
201. Contents loose	Manual
202. Metal ends loose or distorted	Manual
203. Gap between body and cover, max.	Visual
204. Cut or damaged through all impregnated layers	Visual

4.3.2.9 Box, Packing, Drawing C8846788.

Defect	Method of Inspection
CRITICAL	
None defined.	
MAJOR	
None defined.	

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4.3.2.9 (Continued)

	Defects	Method of Inspection
MINOR		
201.	Contents loose	Manual
202.	Hardware missing, loose, broken or improperly engaged	Manual & Visual
203.	Strapping missing, loose, broken or improperly assembled	Manual & Visual
204.	Contents of box exposed	Visual
205.	Box damaged	Visual
206.	Marking missing, incorrect or illegible	Visual
207.	Number of units incorrect	Visual

4.3.3 Inspection testing.- The following tests shall be performed in accordance with the provisions of 4.4.

4.3.3.1 Hardness.- Fifty units from each heat treated lot of components shall be tested for compliance with the values noted on the applicable drawing. Failure of any unit to complywith the drawing requirements or 3.2 shall be cause for rejection of the lot of components.

4.3.3.2 Hydrostatic pressure.- Each chamber and cap sub-assembly shall be subjected to the hydrostatic pressure test. Failure of any sub-assembly to comply with the requirements of 3.3 shall be cause for rejection of the sub-assembly.

4.3.3.3 Shear pin strength.- Fifty pins from each lot of shear pins tested by the contractor shall be tested for shear strength. Failure of any pin to comply withthe requirements of 3.4 shall be cause for rejection of the lot.

4.3.3.4 Protective finish.- The number of units for protective finish tests shall be in accordance with the applicable specifications. Failure of any sample to comply with the requirement of 3.5 and the applicable specification shall be cause for rejection of the lot.

4.3.3.5 Residual magnetism.- Each assembly shall be subjected to the test for excessive residual magnetism. Failure of any assembly to comply with the requirements of 3.6 shall be cause for rejection of the assembly.

4.3.4 Inspection equipment.- The examination and tests shall be made using equipment listed in IEL 10520858.

4.4 Test methods and procedures

4.4.1 Hardness.- Prior to the application ofprotective finish a hardness testing machine shall be used to determine compliance with the hardness requirements specified on the applicable drawing.

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4.4.2 Hydrostatic pressure.- Prior to the application of protective finish, the chamber and cap sub-assembly shall be mounted in the device, using the fixtures for properly sealing all openings, and subjected to the hydrostatic pressure to determine compliance with 3.3.

4.4.3 Shear pin strength.- After application of protective finish, the pin shall be subjected to the shear strength test to determine compliance with 3.4. The test shall be performed in the fixture in conjunction with a tensile testing machine.

4.4.4 Protective finish.- Test procedures for protective finish shall be in accordance with the specification referenced on the component drawing.

4.4.5 Residual Magnetism.- The initiator shall be tested for excessive residual magnetism in an area free of local magnetic effects by placing the initiator 6 inches from and at the same height as the compass, in a north-south horizontal position (cap end at south) with the compass set in an east-west heading.

The initiator shall be moved slowly past the compass in longitudinal direction for its entire length, rotated 90 degrees in the horizontal position about its longitudinal axis and moved slowly back past the compass. This procedure shall be repeated for each 90 degree rotation of the initiator.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking.- Packing and marking shall be in accordance with Drawings C8846787 and C8846788.

5.2 Data Cards.- Data card information shall be as specified in MIL-STD-1167.

6. NOTES

6.1 Ordering data.- Procurement documents should specify the title, number and date of this specification.

6.2 A mechanical failure is defined as any deformation or breakage of a part, the occurrence of which is other than a design function.

6.3 Supersession data.- This specification includes the requirements of Purchase Description FA-PD-MI-2492 Rev. 0 dated 24 March 1961.

Custodian:

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Preparing Activity:

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